Model selection for forecasting the US dollar to NOK currency exchange rate

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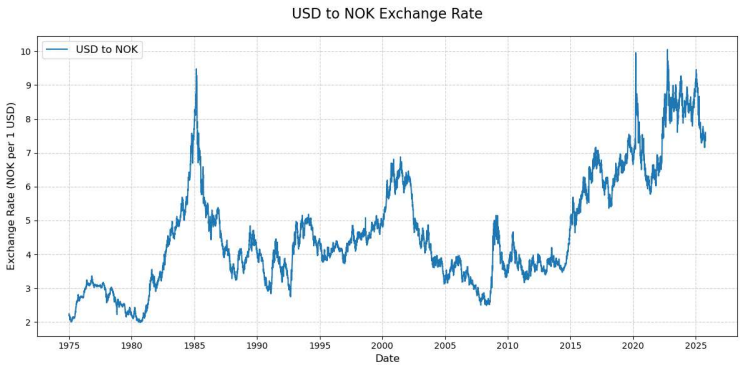


The graph represents the USD to NOK exchange rate from the 2nd of January 1975 until the 13th of October 2025. The dataset contains multiple exchange rates that share a common reference currency. Almost all the daily exchange rates are provided, and we remove the row for missing values. To calculate the exchange rate "US dollar to NOK currency" we operate the calculus: Serie = data["XUDLNKD"]/data["XUDLUSS"] on the provided dataset "boe_fx.csv". Then, the data set is split in a training set and a testing set containing the 200 last values.

GOAL: Find the best model to predict the next day's exchange rate. We chose to work with three kinds of models: ARIMA, GARCH and ARIMA-GARCH.

APPLICATION: Applications are diverse. The model predicts the next day's exchange rate and its volatility, allowing traders to buy or sell and avoid high-risk (high volatility) periods to maximize their returns. Such a model allows you to buy currency at a better rate by intelligently choosing the day of purchase.

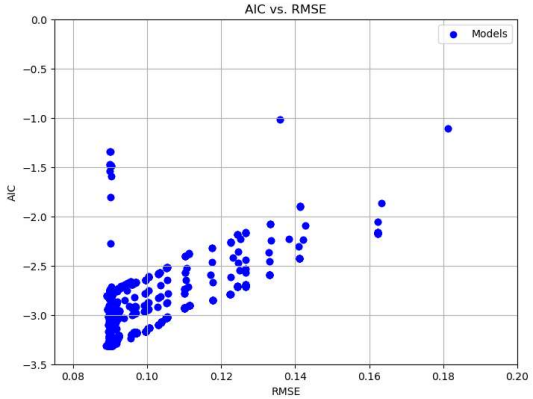
Evaluation Metrics:

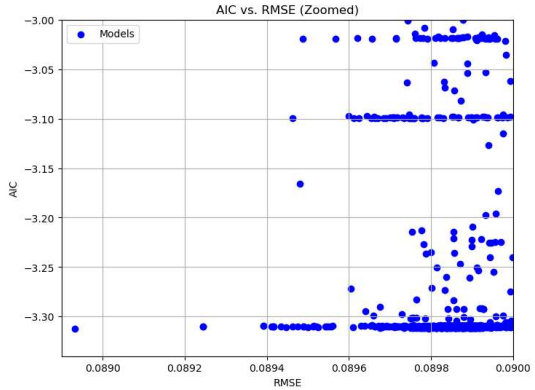
1. **AIC (Akaike Information Criterion):** Measures the **relative quality** of a model by balancing goodness-of-fit and complexity.
2. **RMSE (Root Mean Square Error):** Quantifies the **average magnitude** of prediction errors.

Model	ARIMA	GARCH	ARIMA-CARCH
Description	Captures the **conditional mean** of a time series using autoregression, differencing, and moving averages.	Captures **conditional variance** by tracking how past squared errors and volatility influence future uncertainty. (applied on the relative gain r)	Jointly estimates **both the conditional mean (ARIMA) and conditional variance (GARCH)** for a complete time series forecast.
Parameters	**p** – autoregressive order **d** – differencing order **q** – moving-average order	**p** – number of past variances **q** – number of past squared residuals	**pm, dm, qm** – ARIMA mean parameters **pv, qv** – GARCH variance parameters
Best model for AIC	ARIMA(p=4,d=1,q=4)	GARCH(p=3,q=3)	ARIMA(6,0,6)-GARCH(3,3)
Best AIC value	-35463	-6.25	-3.31
Best model for RMSE	ARIMA(6,0,7)	GARCH(3,0)	ARIMA(5,0,5)-GARCH(2,3)
RMSE value	RMSE = 0.089382	RMSE = 0.089808	RMSE = 0.088933
Relative RMSE (%)	-0.467	0.008	-0.967

Relative RMSE (%): It is the relative RMSE value compared to the RMSE of the dumb prediction x_pred(t+1) = x(t) (RMSE = 0.089801) A positive value means that the model do worse prediction (greater RMSE). We are looking for the lowest relative RMSE.

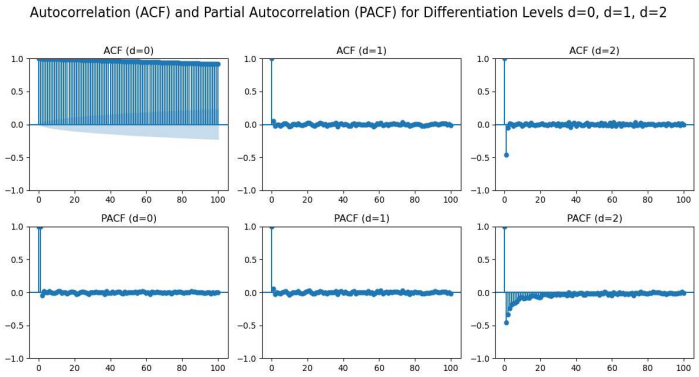
Test parameter range : dm: 0:3 , pm: 0:7 , qm: 0:7 , pv: 0:3 , qv: 0:3





Consistency: The RMSE measures how far the model's predictions are from the actual data, so it is the primary metric to follow when evaluating performance. Once several models achieve similarly low RMSE values, the AIC becomes useful to refine the choice: among those good models, we can select the one with the lowest AIC (even if its RMSE is slightly higher) to favor a simpler and smoother model. In the graph, one point stands out clearly in the bottom left corner— the model ARIMA(5,0,5)-GARCH(2,3) . It has the best RMSE of all and an AIC value that is just as good as the other top models. This confirms that AIC alone is not sufficient to judge model quality: the model with the best AIC did not produce strong predictive performance (RMSE = 0.0897), while RMSE provides a more direct and reliable measure of accuracy. However, when RMSE differences are very small, the AIC is helpful to choose the smoother, more parsimonious model.

Best model : ARIMA(5,0,5)-GARCH(2,3) based on RMSE and AIC



For **d = 0**, the ACF shows very high positive correlations that decay very slowly, remaining close to one across many lags. The PACF, on the other hand, has one dominant spike at lag 1 followed by small, insignificant values. This combination indicates that the original series is non-stationary, probably containing a strong trend or a unit root component such as a random walk.

For **d = 1**, the ACF drops quickly after lag 0, and most of the autocorrelations are close to zero. The PACF also shows only one large initial spike, with subsequent values hovering around zero. This pattern suggests that after the first differencing, the series has become stationary — the mean and variance appear stable.

For **d = 2**, the ACF stays near zero with a small negative spike at the first lag, while the PACF displays a mild oscillating pattern with small alternating negative values. These features point to over-differencing: the series was already stationary after one differencing, and applying a second difference has introduced unnecessary short-term negative correlation. Thus, one differencing step (d = 1) is sufficient.

Remarks : Compare the AIC values between ARIMA and GARCH or ARIMA-GARCH is not relevant because it has a different scale (residuals vs. differentiated data

Libraries used : forecast (for ARIMA) and rugarch (for GARCH and ARIMA-GARCH)

Softwares used : R to find the best models and Python to plot graphs.